

Introduction to Virtualization

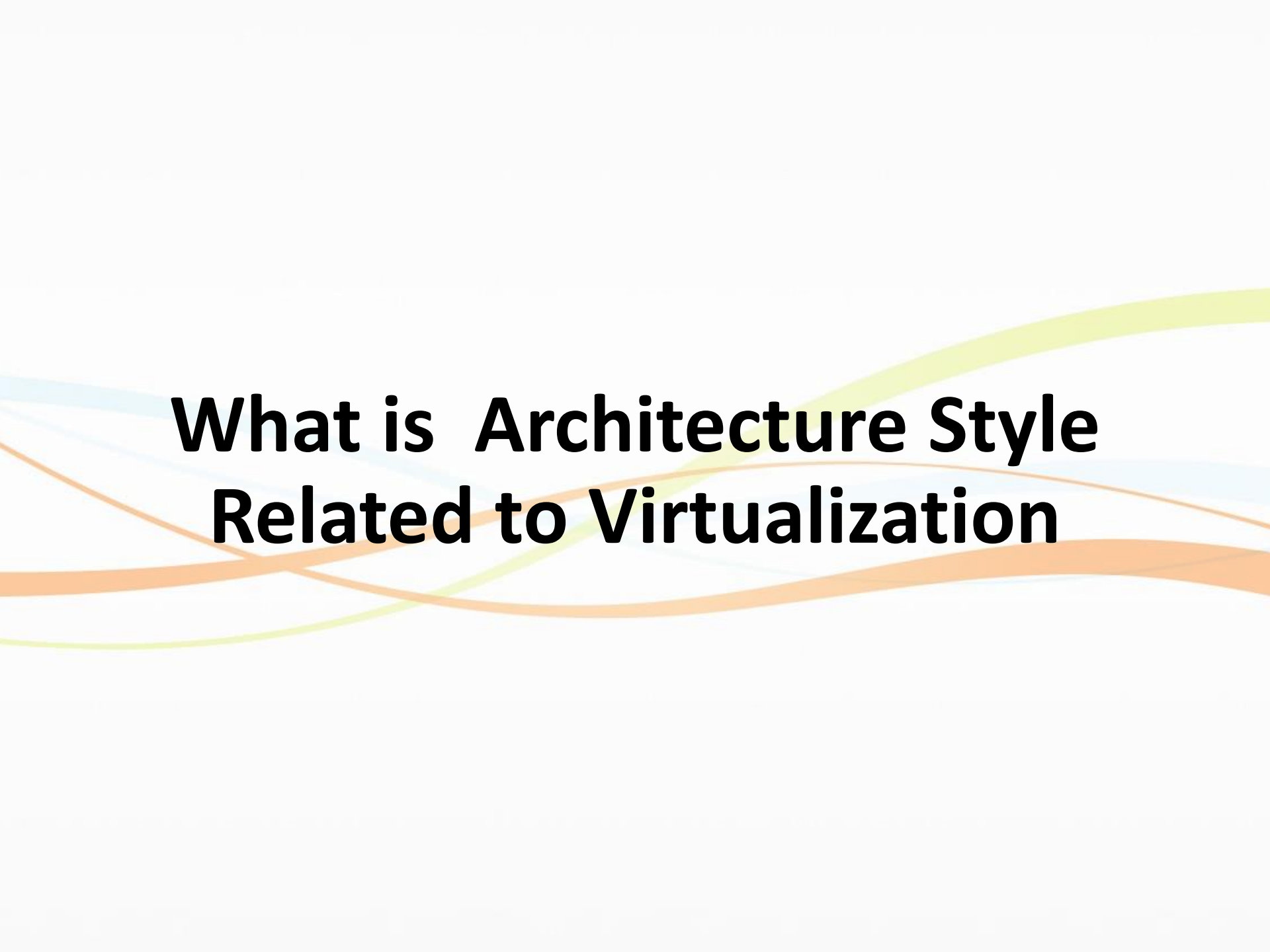


Contents

- ▶ What is Architecture Styles Related to Virtualization ?
 - ▶ Implementation Levels of Virtualization
 - ▶ What is Hardware Virtualization ?
 - ▶ VM and VMM(=Hypervisor)
 - ▶ History of Virtualization
 - ▶ Benefits of Virtualization
 - ▶ Usages of Virtualization
- 

What is Virtualization Concept?

- ▶ Virtualization, in computing, is
 - ❖ the creation of a virtual (rather than actual) version of something,
 - ❖ such as a hardware platform, operating system, a storage device or network resources
- ▶ Hardware Virtualization basically
 - ❖ allows one computer to do the job of multiple computers,
 - ❖ by sharing the resources of a single hardware across multiple environments



What is Architecture Style Related to Virtualization

A fact sheet of architecture styles

	<u>Time Economy</u>	<u>Space Economy</u>	Complexities	<u>Security</u>	Interoperability	<u>Hardware Indep.</u>	<u>Software Indep.</u>	<u>Installability</u>	<u>Reliability</u>	<u>Error-tolerance</u>	<u>Availability</u>	Understand.	User Interface	<u>Learnability</u>
OO	+	+	+				+		+				+	-
Batch sequential					-				-					+
Pipeline & Filter		-			-				+					+
Process Control	+					-								-
Repository	+	+							+			-		-
Blackboard	-	+							+					-
Mainprog-Subroutine	+	+		-	-				-			-		-
MasterSlave	+								-					-
Layered	-			+		+			+	+	+	+		-
Virtual Machine	-	-		+		++	++	+	+	+	+			-
EventBased (non-buf)					+							+		+
MsgPassing (buffered)		-			+							+		+
MVC					+									+
PVC					+									+
ClientServer														+
Multi-tier	-	-		+	+									+
Broker	-													-
Service Ori. Arch. (SOA)	-	-			++	++	++	+	++		+	+	+	++
Component Based					++		+		++		+			++

Virtual Machine Architecture Style

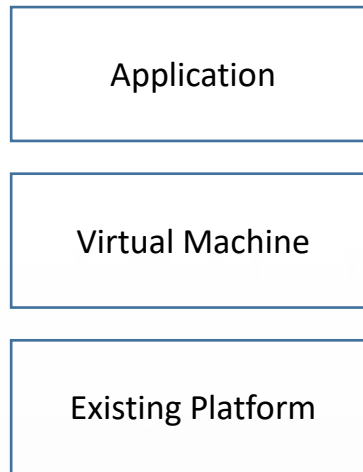
► Synopsis

- ❖ A virtual machine is built up on an existing system and provides a virtual abstraction, a set of attributes, and operations.
- ❖ In most cases a virtual machine separates a programming language or application environment from an execution platform.
- ❖ A virtual machine may appear similar to emulation software.

Virtual Machine Architecture Style

▶ Structure

- ❖ Divide a system into three parts: existing platform, virtual machine and application

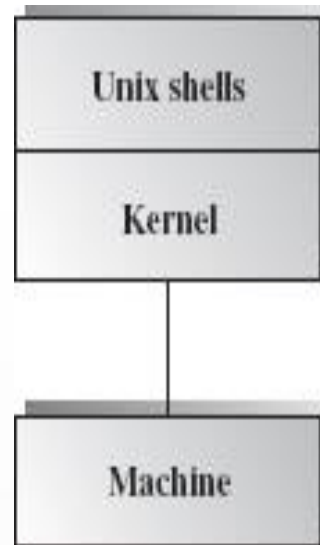


Virtual Machine: Example

▶ Application

❖ Unix OS' Shell Interpreter

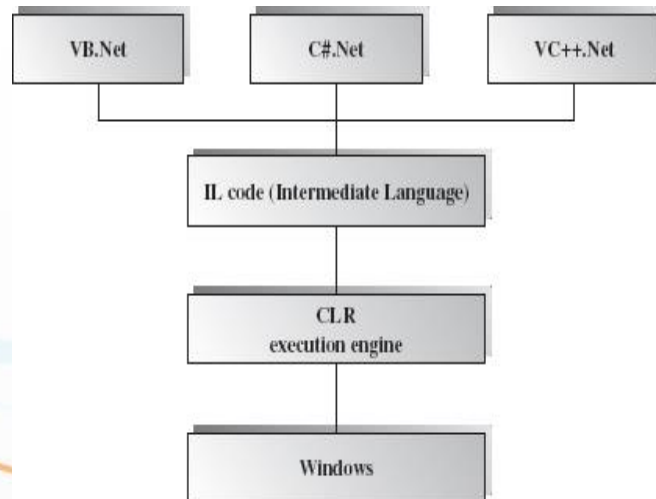
- provides multiple shells such as C shell, Korn shell, and Born shell on top of the Unix kernel.



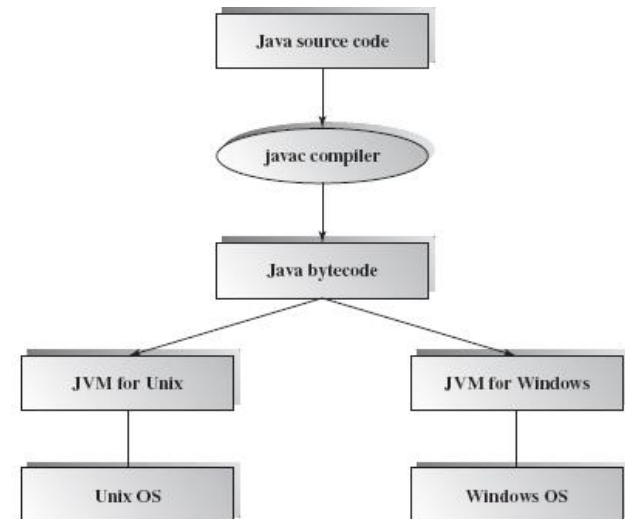
Virtual Machine: Example

❖ Language Runtime Environment Platform

- Common Language Runtime (CLR) of MS .NET platform
 - plays the role of a virtual machine that uses a single intermediate language to unify several modules VB.NET, VC.NET, and C#



- Java Virtual Machine (JVM)
 - The Java bytecode and other Java internal code generated by a compilation system (e.g., javac) can run on any operating system that supports JVM



Virtual Machine: Example

► Android Platform



Virtual Machine: Benefits & Limitations

▶ Benefits

- ❖ Portability and machine platform independency
- ❖ Simplicity of software development
- ❖ Simulation for disaster working model

▶ Limitations

- ❖ Slow execution of the interpreter due to the interpreter nature
- ❖ Additional overhead due to the new layer

▶ Related architecture:

- ❖ Interpreter, repository, and layered architecture



Implementation Levels of Virtualization

Implementation Level of Virtualization

► Level of Implementations

- ❖ Application
- ❖ Library Support
- ❖ Operating System
- ❖ Hardware Abstraction
- ❖ Instruction Set Architecture

Application level

JVM / .NET CLR / Panot

Library (user-level API) level

WINE/ WABI/ LxRun / Visual MainWin / vCUDA

Operating system level

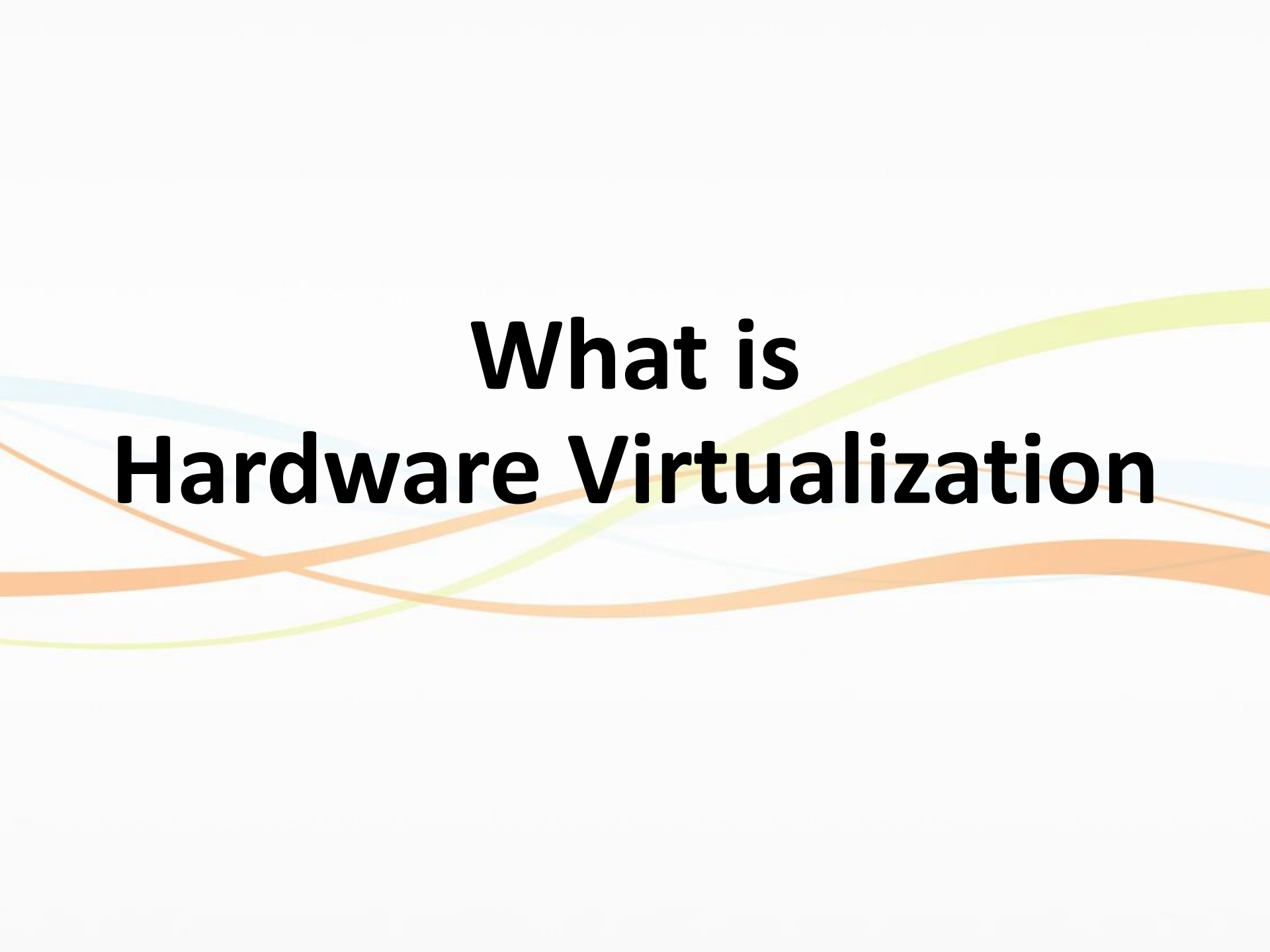
Jail / Virtual Environment / Ensime's VPS / FVM
LXC, Docker

Hardware abstraction layer (HAL) level

VMware / Virtual PC / Denali / Xen / L4 /
Plex 86 / User mode Linux / Cooperative Linux

Instruction set architecture (ISA) level

Bochs / Crusoe / QEMU / BIRD / Dynamo



What is Hardware Virtualization

What is Hardware Virtualization

- ▶ Fundamental idea – abstract hardware of a single computer into several different execution environments
 - ❖ Similar to layered approach
 - ❖ The layer creates virtual system (**virtual machine**, or **VM**) on which operation systems or applications can run

What is Hardware Virtualization

- ▶ **Hardware virtualization(or platform virtualization)** refers to the creation of a virtual machine that **acts like a real computer with an operating system**, so that software executed on these virtual machines is separated from the underlying hardware resources.
- ❖ For example, a computer that is running Microsoft Windows may host a virtual machine that looks like a computer with the Ubuntu Linux operating system; Ubuntu-based software can be run on the virtual machine.

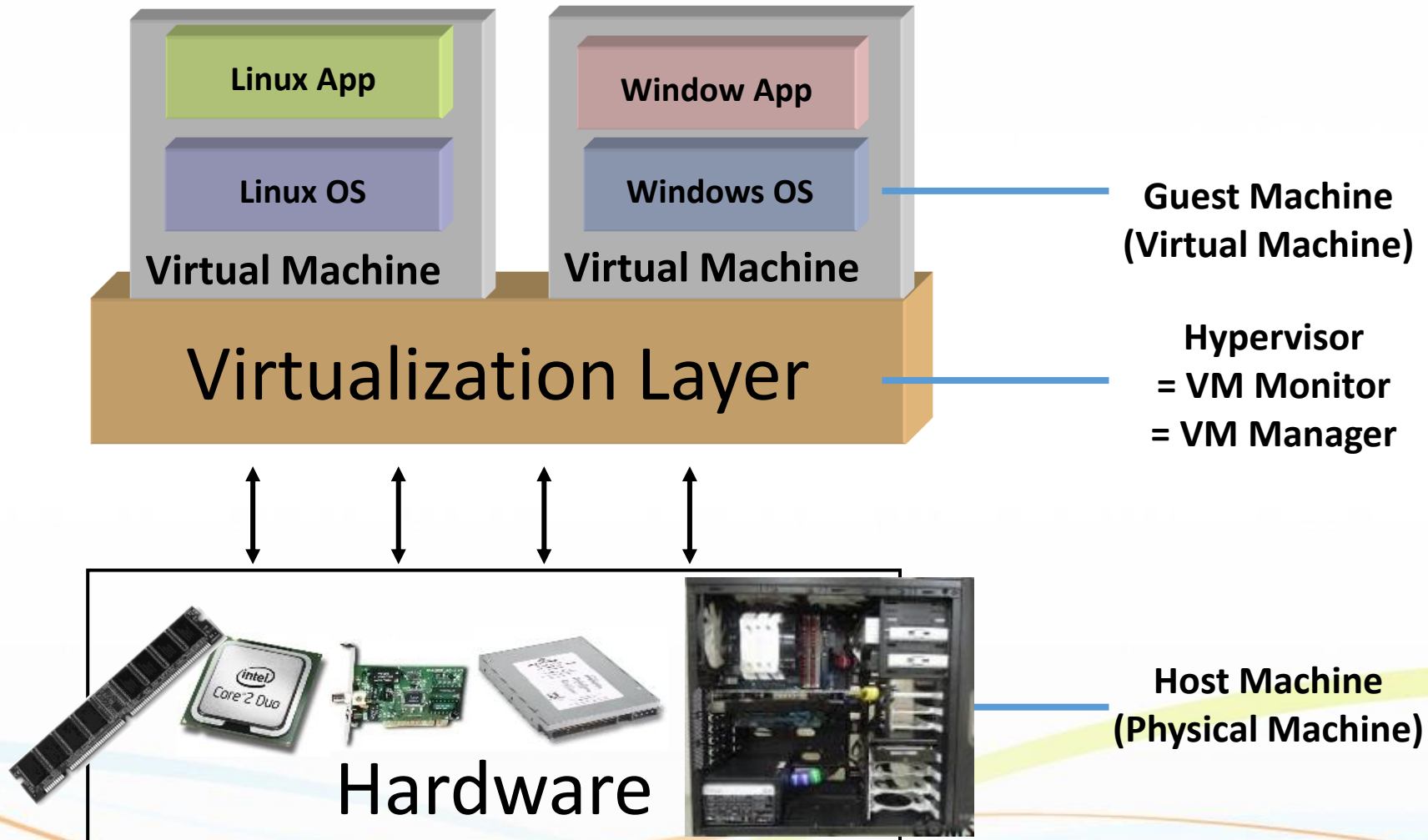
What is Hardware Virtualization

- ▶ If virtualization is defined as enabling multiple operating systems to run on a single host computer, then **the essential component** in the virtualization stack **is the hypervisor**.
- ▶ **Hypervisor**, also called **Virtual Machine Monitor/Manager (VMM)**, creates a virtual platform on the host computer, on top of which multiple guest operating systems are executed and monitored.

What is Hardware Virtualization

- ▶ Several components
 - ❖ **Host** – underlying hardware system
 - ❖ **Virtual machine manager (VMM)** or **hypervisor** – creates and runs virtual machines by providing interface that is *identical* to the host
 - (Except in the case of paravirtualization)
 - ❖ **Guest** – process provided with virtual copy of the host
 - Usually an operating system
- ▶ Single physical machine can run multiple operating systems concurrently, each in its own virtual machine

Hardware Virtualization (Terms)



Virtual Machines

- ▶ Virtual machine (VM) is
 - ❖ **a software implementation of a machine** (computer) that executes programs like a real machine
- ▶ Virtual machines provide
 - ❖ **Hardware independence**
Guest VM sees the same hardware regardless of the host hardware
 - ❖ **Isolation**
VM's operating system is isolated from the host operating system
 - ❖ **Encapsulation**
Entire VM encapsulated into several files contained in a single directory

Virtual Machine Monitor or Hypervisor

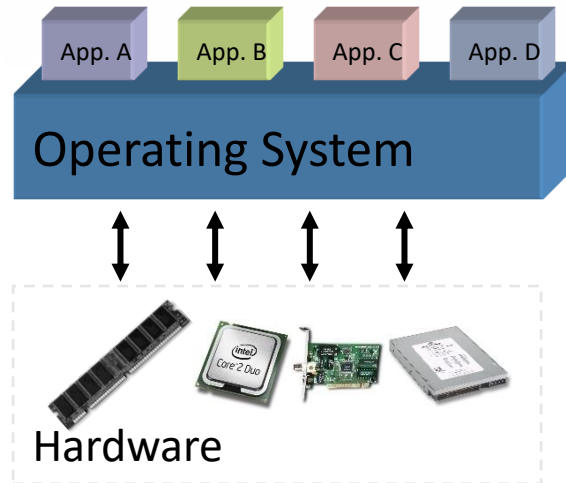
▶ VM Monitor(VMM) or Hypervisor

- ❖ presents a **SW interface to guest software**,
- ❖ isolates state of guests, and
- ❖ **protects itself from guest software** (including guest OS)

▶ VMM must be **at a higher privilege level than guest VM**, which generally run in user mode

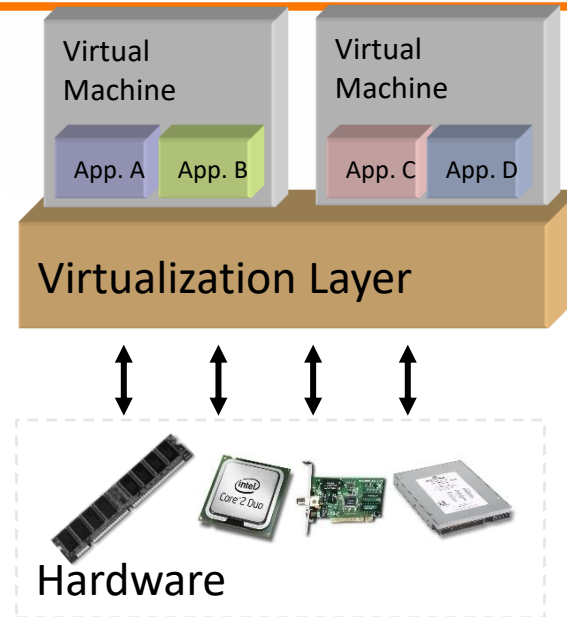
- ❖ Guest software should behave on a VM exactly as if running on the native HW
- ❖ Guest software should not be able to change allocation of real system resources directly
- ❖ Hence, VMM must control everything even though guest VM and OS currently running is temporarily using them

Non-Virtualized vs. Virtualized



'Non-virtualized' system

- A **single OS controls all hardware** platform resources
- Running multiple applications on same machine often creates conflict
- Inflexible and costly infrastructure



Virtualized system

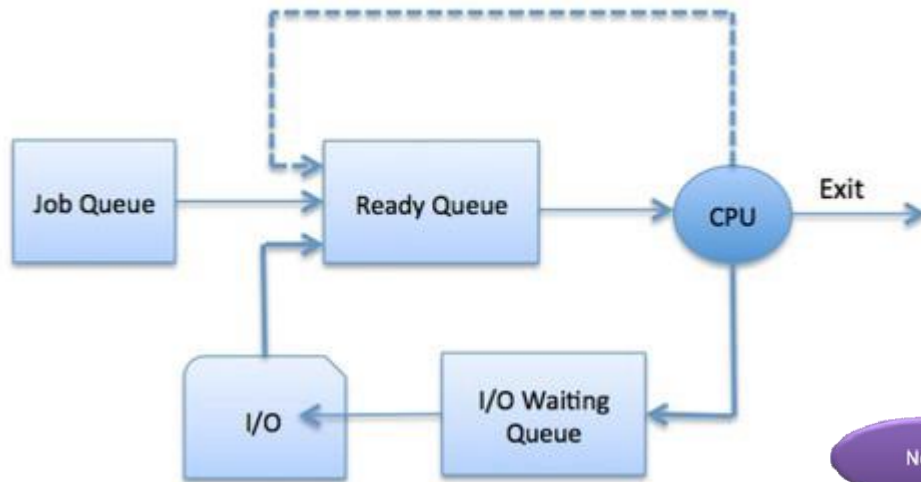
- It makes it possible to **run multiple Virtual Machines on a single physical platform**
- **Hardware-independence** of operating system and applications
- Virtual machines **can be provisioned** to any system



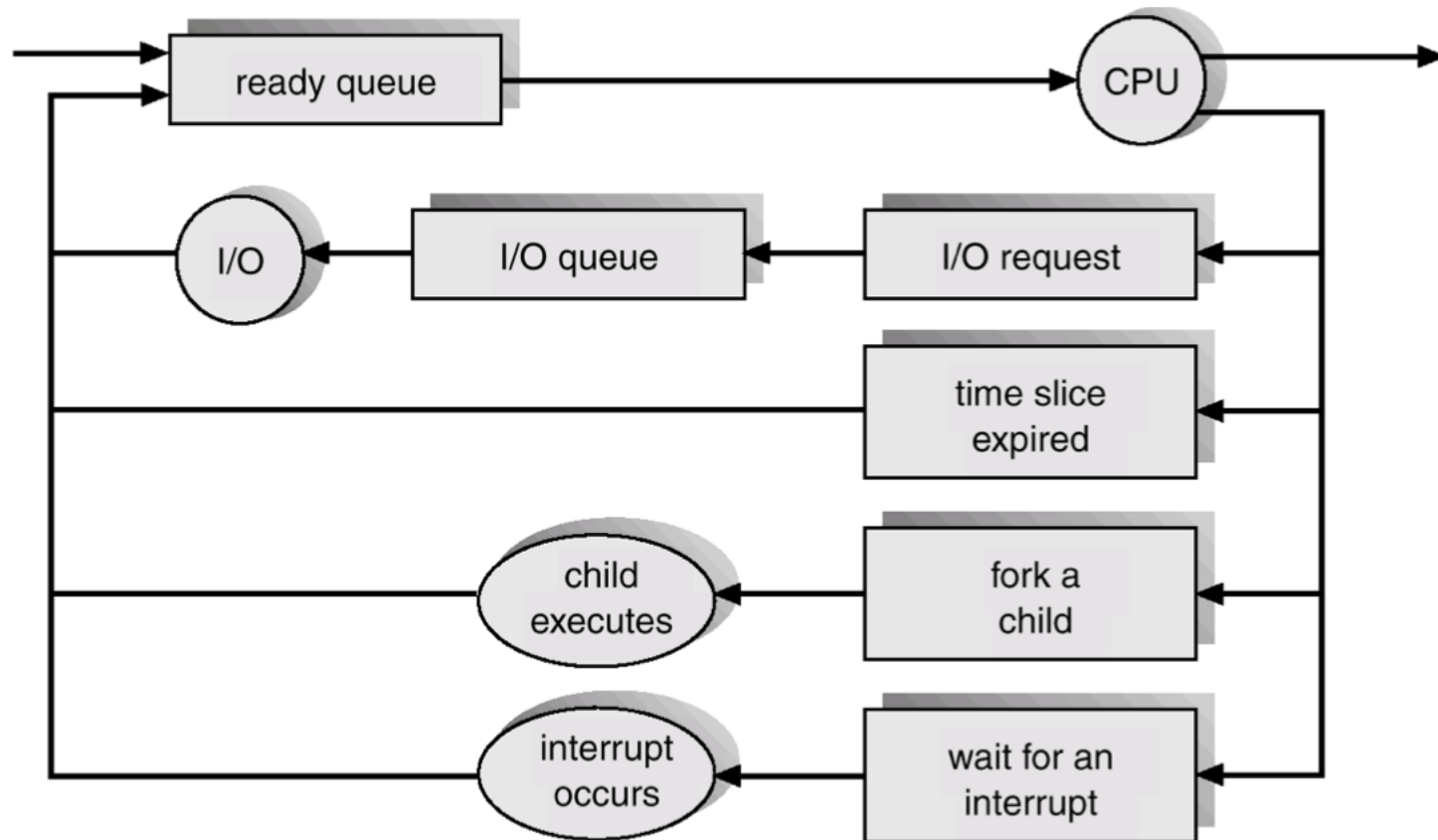
Core Techs of Hypervisor for Hardware Virtualization

CPU Management

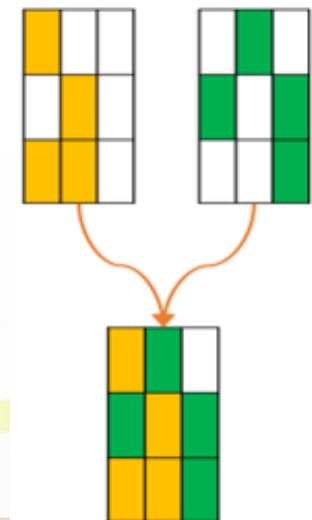
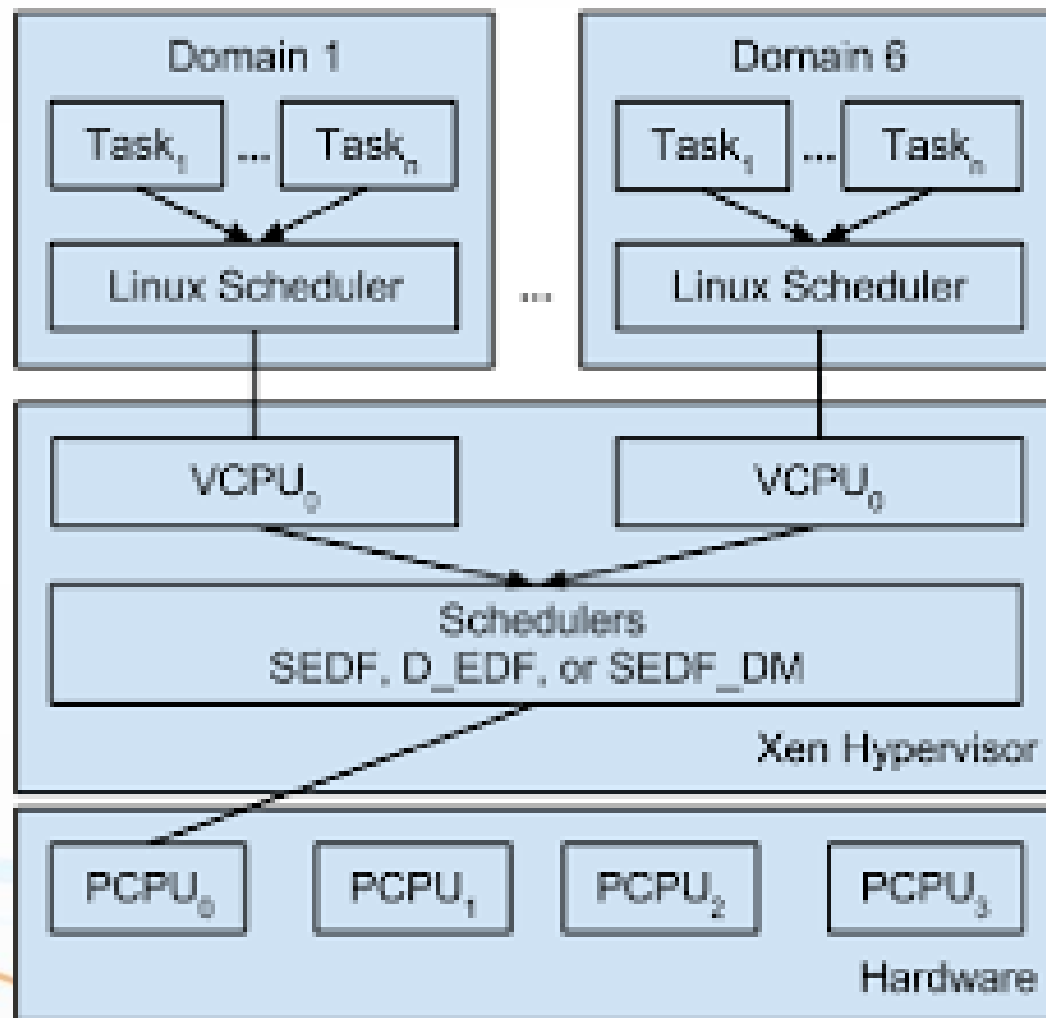
CPU Scheduling for Time Sharing



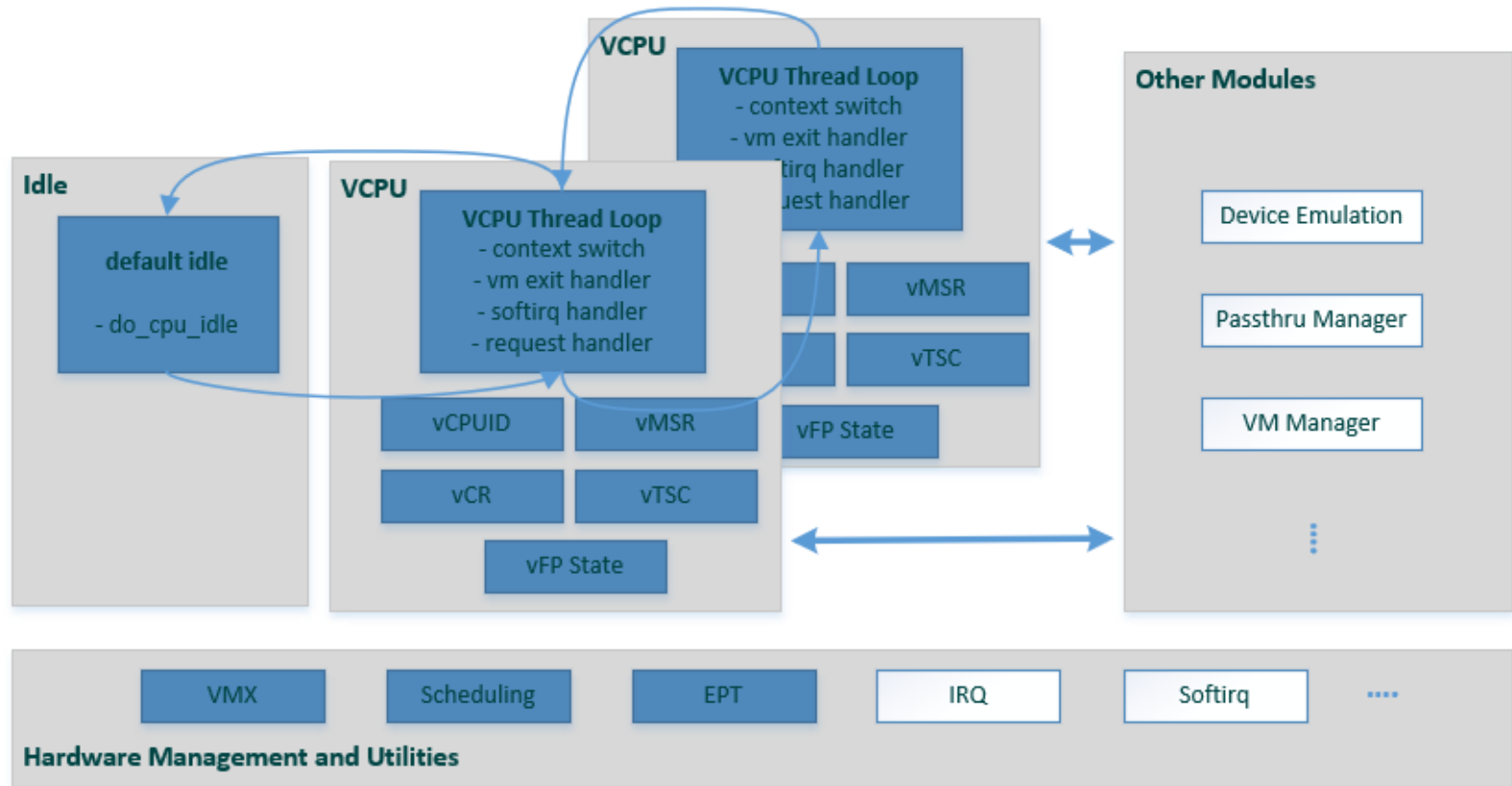
CPU Scheduling



CPU Virtualization / CPU Scheduling

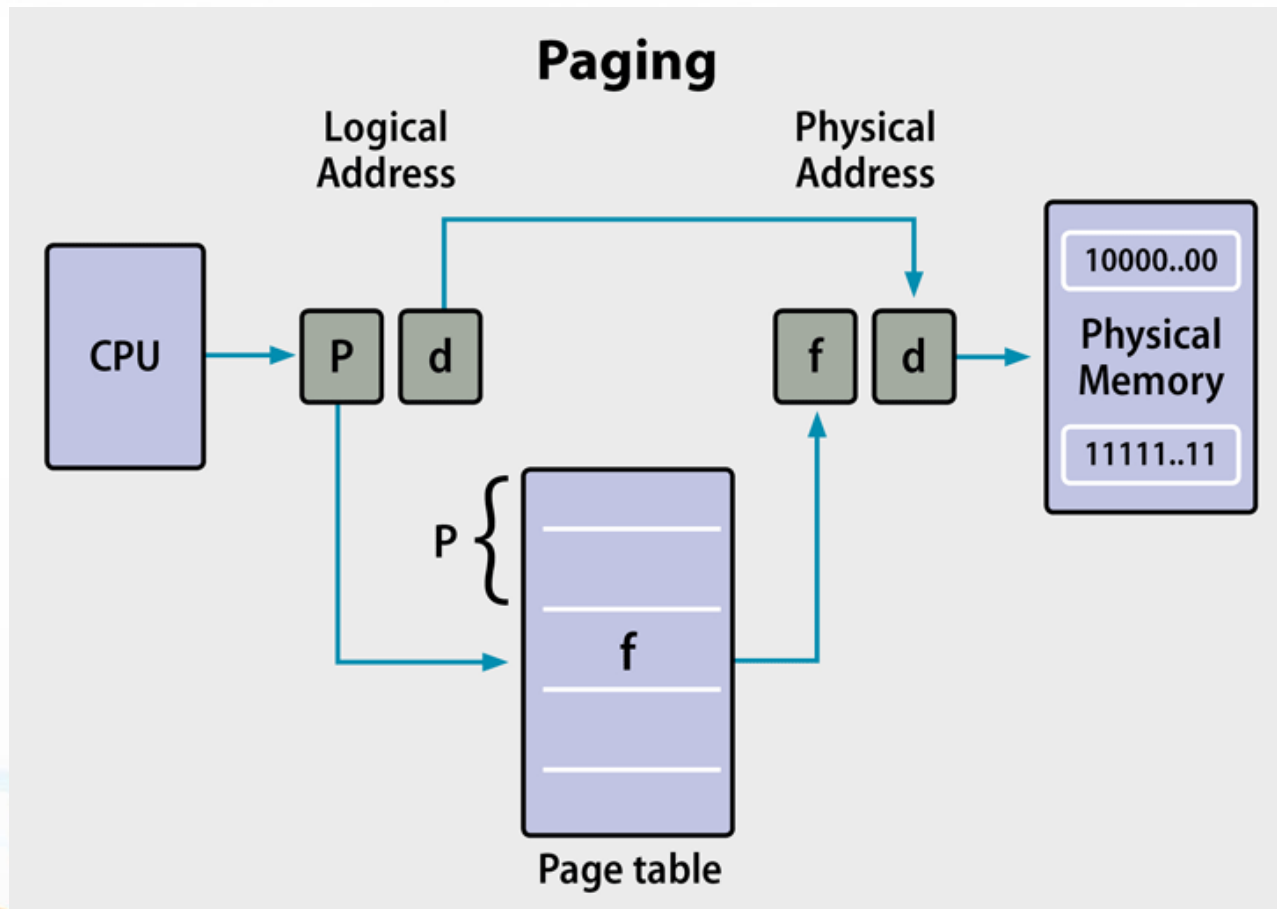


CPU Virtualization

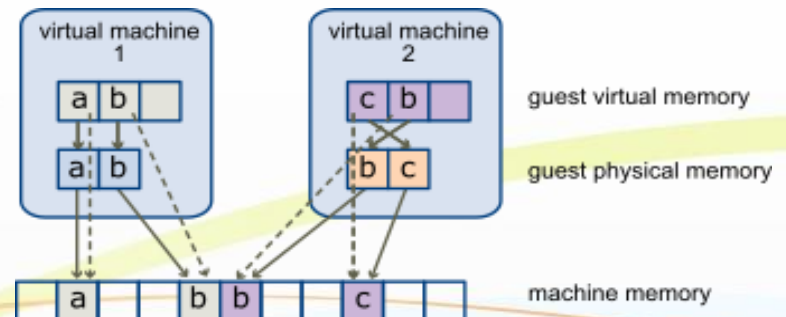
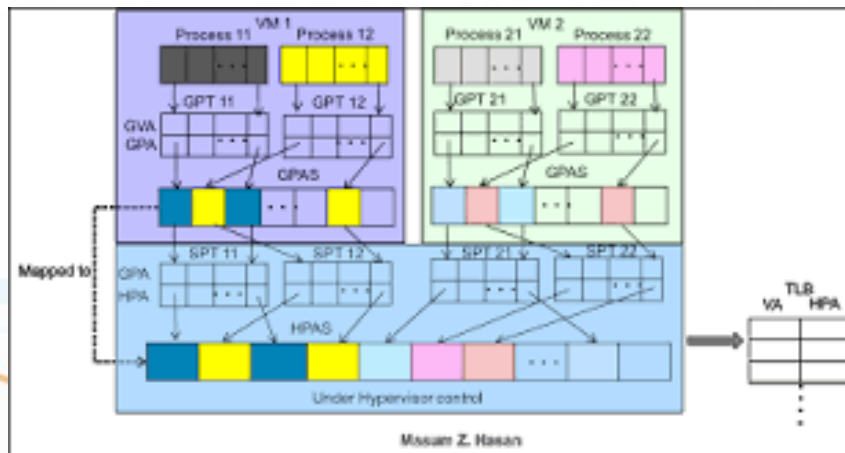
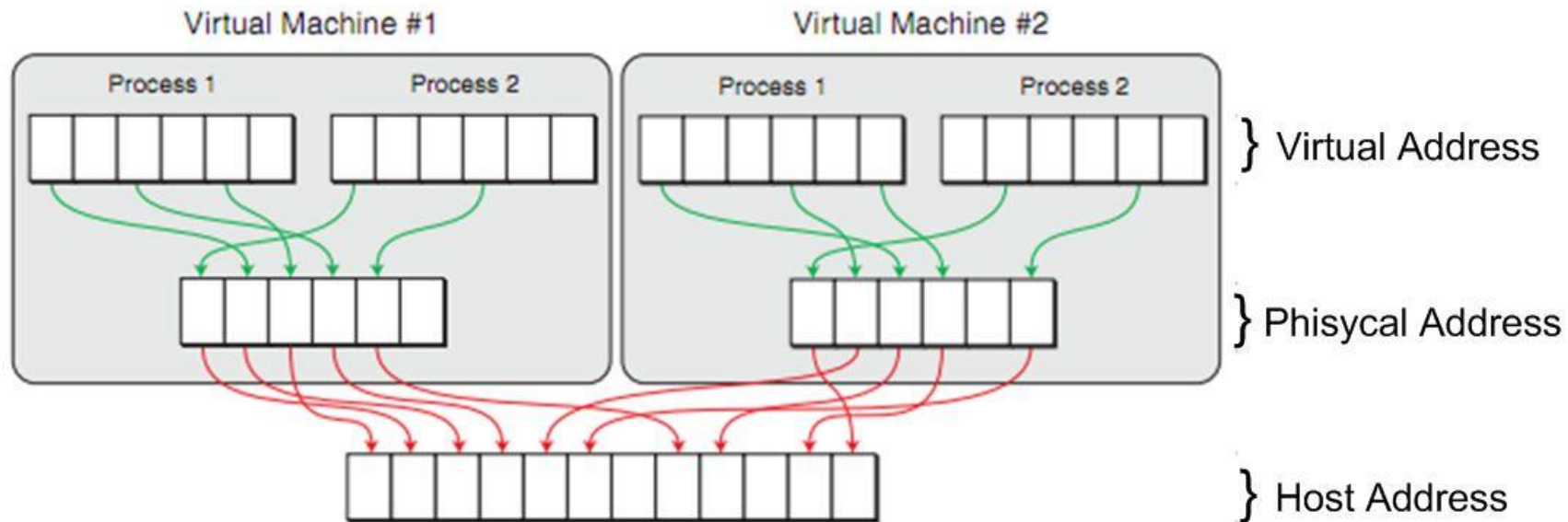


Memory Virtualization

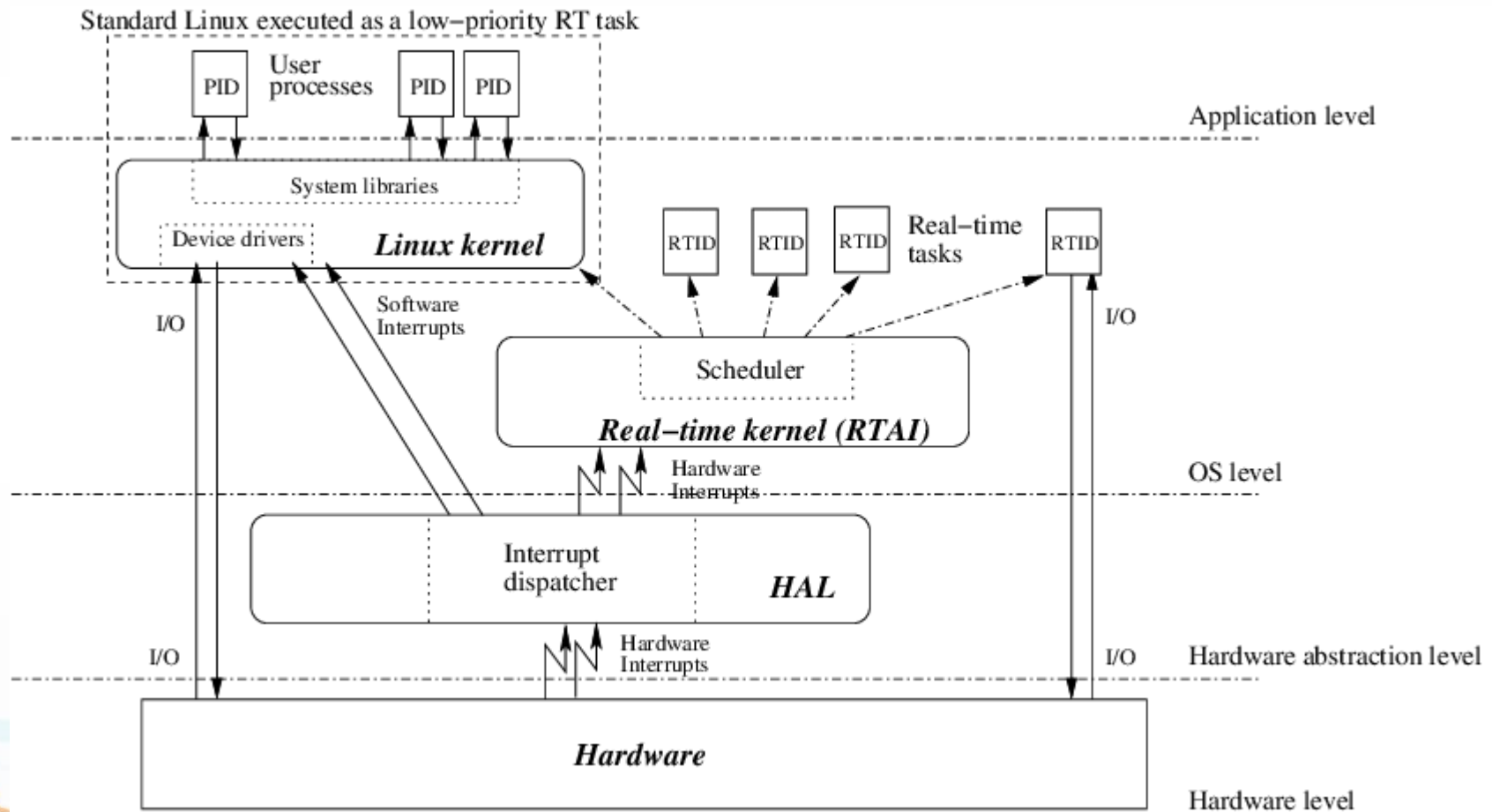
Translation Table for Address Mapping



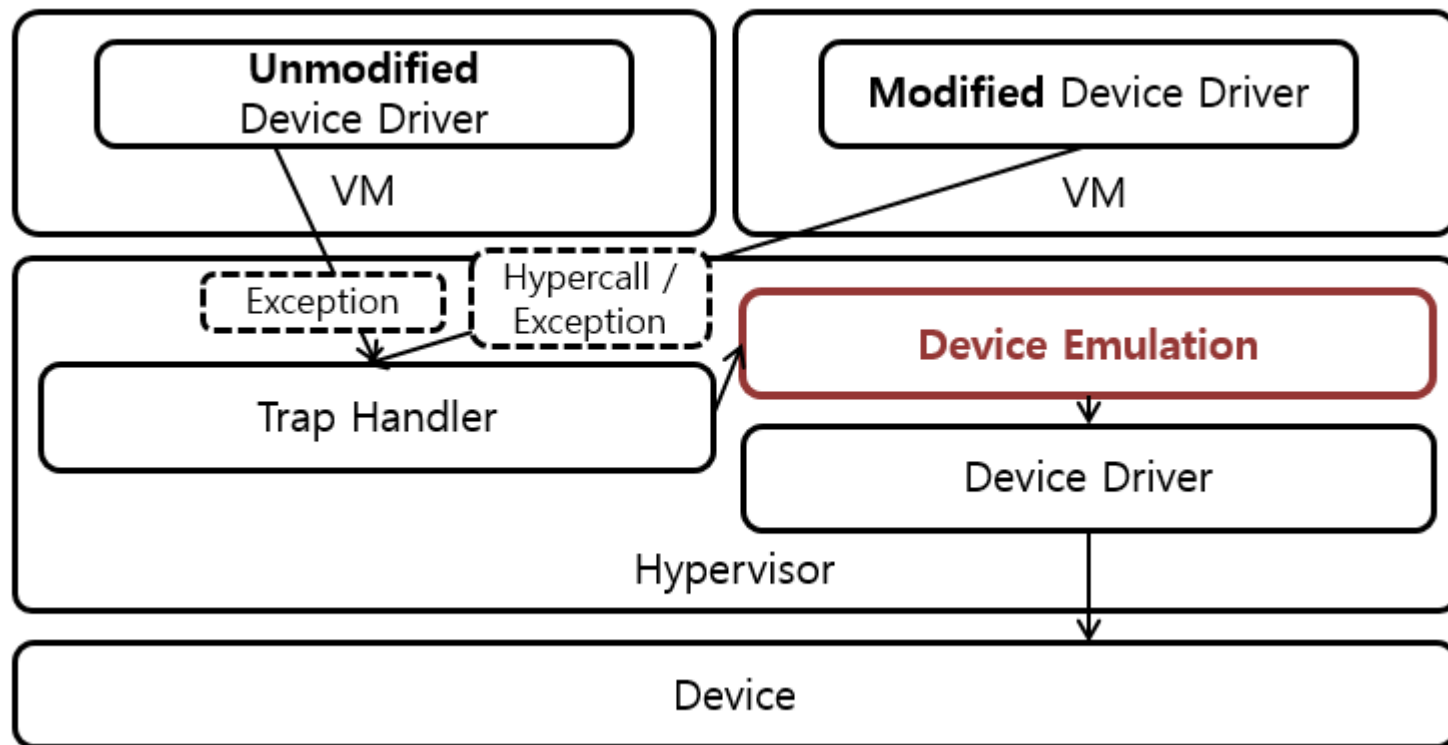
Memory Virtualization



I/O Management



I/O Virtualization

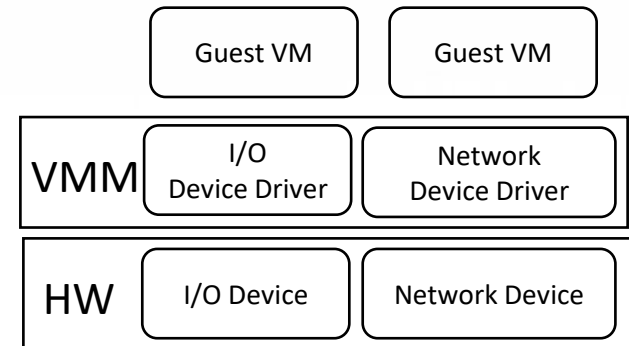


I/O Virtualization

▶ 2 Ways of I/O Virtualization

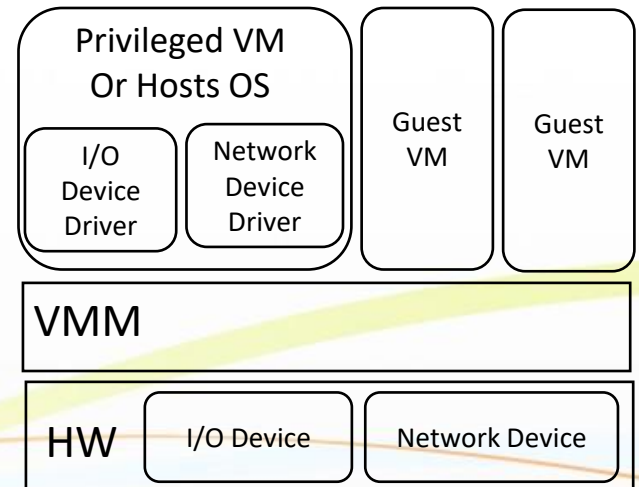
❖ I/O Virtualization in VMM

- Rewritten Device Driver in VMM
- + High Performance
- - High Engineering cost
- - Low Fault tolerance (driver bugs)



❖ Hosted I/O Virtualization

- Existing device drivers in a host OS
- + Low Engineering cost
- + High fault tolerance
- - Performance Overheads



VMM Overhead

- ▶ Depends on the workload type
 - ❖ CPU-bound workload
 - => zero-virtualization overhead
 - Runs at native speeds since OS rarely invoked
 - ❖ IO-intensive workload
 - => high virtualization overhead
 - Execute many system function calls and privileged instructions
 - ❖ IO-intensive workload, but IO-bound
 - => Low virtualization overhead
 - Low processor utilization since waiting for IO
 - Processor virtualization can be hidden



History of Virtualization

1960 ~ Middle of 1990

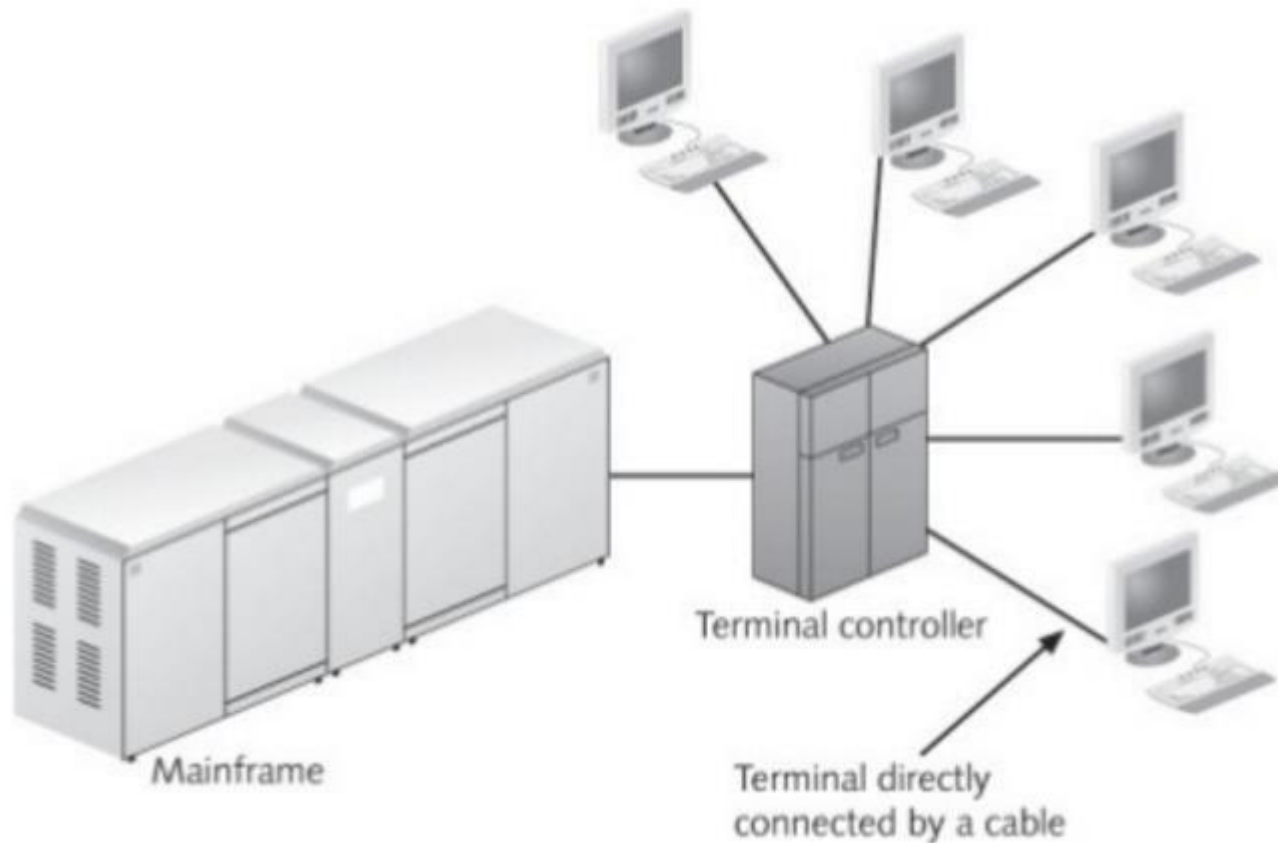
▶ 1960s-1970s (Begin)

- ❖ Because of high cost of computing machines, sharing resources is the big issue
- ❖ Virtualization began in the 1960s, as a method of logically dividing the system resources provided by [mainframe computers](#) between different applications.
- ❖ IBM VM/370: A VMM for IBM mainframe

▶ 1980s-1990s (Lost Attention)

- ❖ Since PC era (1980s)
- ❖ Because of Cheap HW, no need to share hardware
- ❖ HW did not support virtualization well

Mainframe Virtualization in 1960



Time-sharing mainframe with terminals

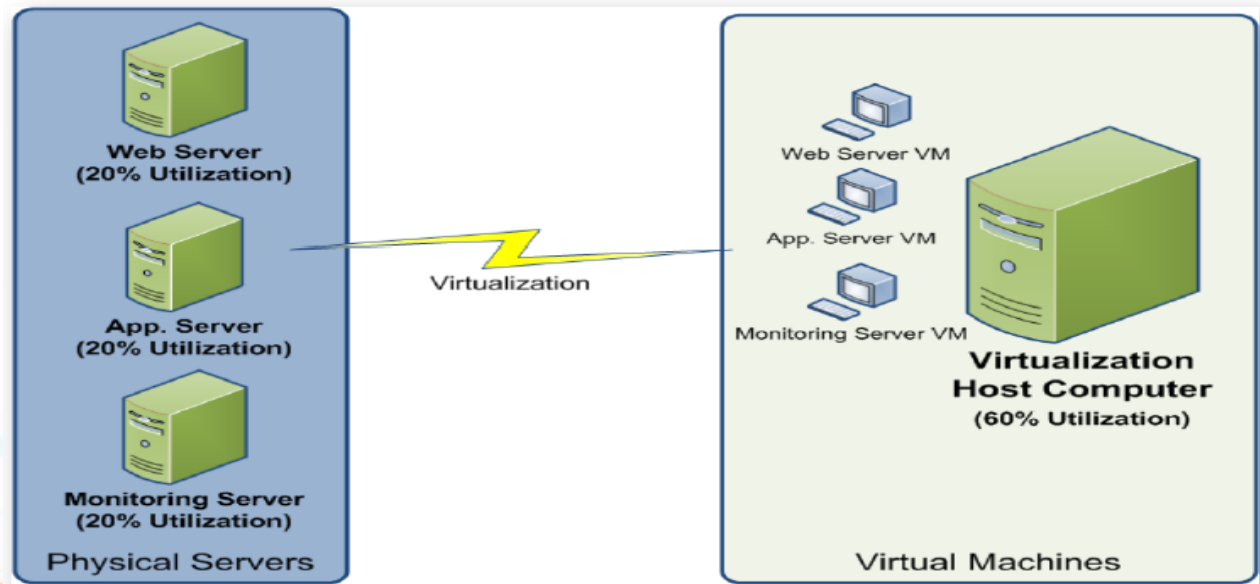
The late 1990s (Re-Attention)

- ▶ Because of **Server Proliferation Problem**

=> **Server Consolidation**

❖ Toward cost-effective large-scale computing

- ▶ Stanford's research project("Disco") team founded VMware in 1998



1998 ~ 2003: VMware, Xen

▶ 1998-2002: VMware

- ❖ **Full-virtualization:** running unmodified OS
 - Starting with SW-based binary translation
- ❖ Success in industry and academia

▶ 2003: Xen

- ❖ University of Cambridge's project (**Open source**)
- ❖ **Para-virtualization:** Modified OS for near-native performance
- ❖ 2003~ Lots of research started based on Xen
- ❖ 2007: Acquired by Citrix

2005 ~ Now: Become a Mainstream

▶ 2005-2006: **HW-assisted virtualization**

❖ x86 virtualization

- Intel VT-x , AMD-V
- Running unmodified OS with near-native performance
- Default functionality now

▶ 2006 ~ Now

- ❖ **Cloud computing: Infrastructure-as-a-Service (IaaS)**
- ❖ **Virtual desktop infrastructure (VDI)**
- ❖ **Mobile virtualization**

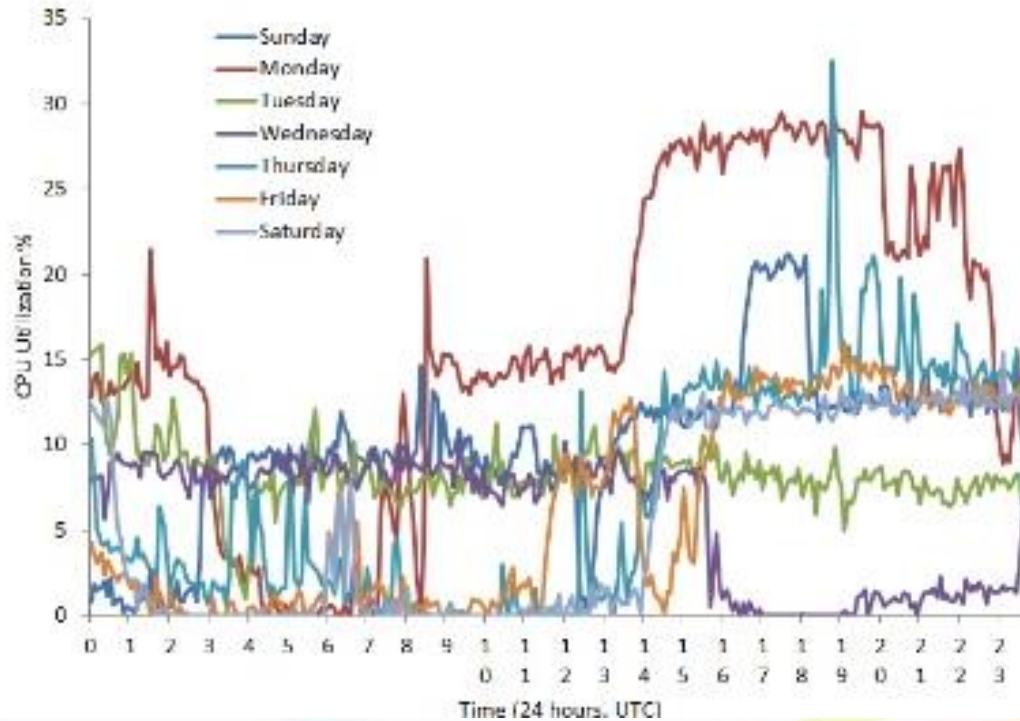


Why Virtualization?

Benefits of Virtualization

► Consolidation for resource utilization optimization

❖ Low resource utilization of each server machine



**7.3%
Average**

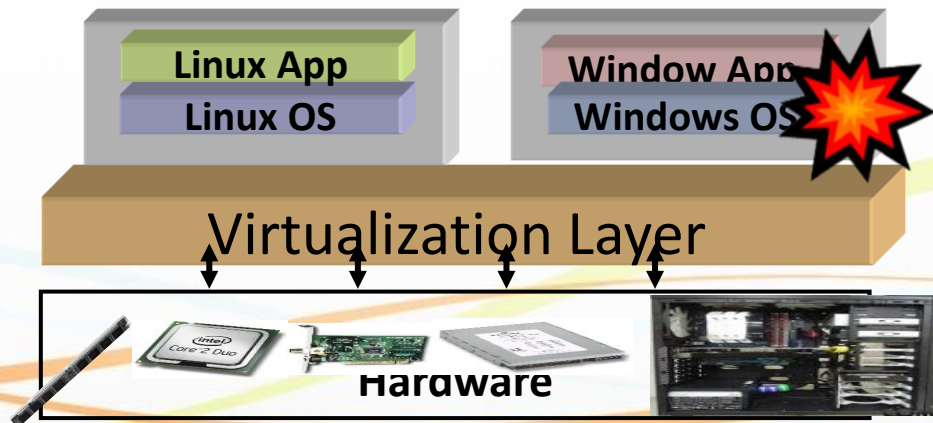
"Host server CPU
utilization in Amazon EC2
cloud," Huan Liu's Blog,
2012

=> Low total cost of ownership (TCO): Infra and Energy

Benefits of Virtualization

► Co-Existence, but Strong isolation of VMs

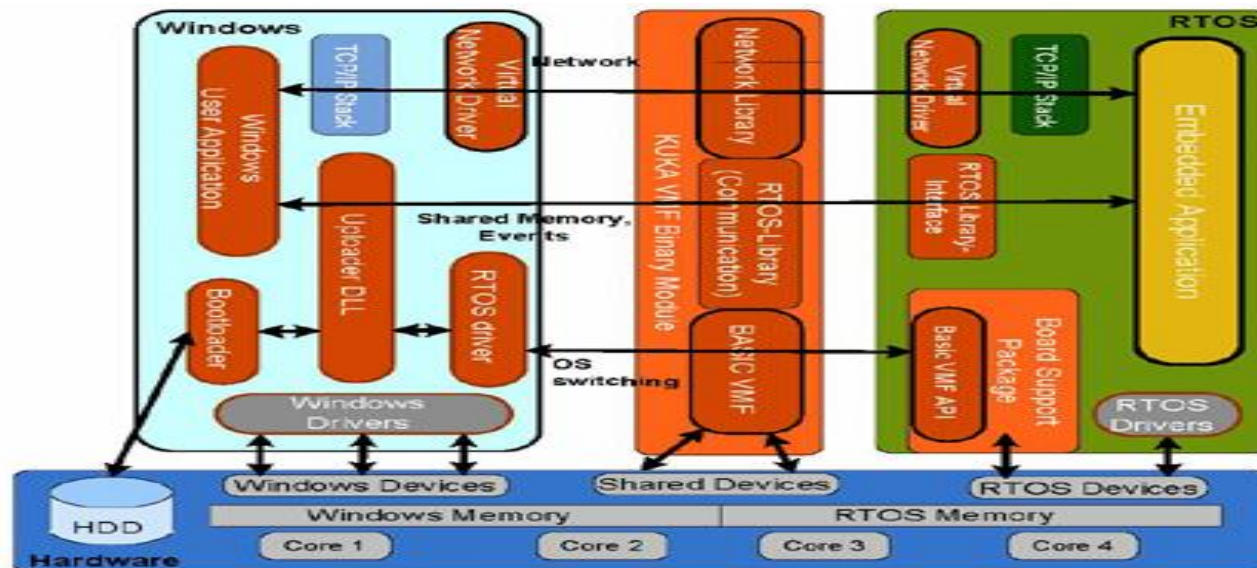
- ❖ Separate between VMs, Separate VM from Host
- ❖ Better isolation of (catastrophic) failures, security leaks
- ❖ Good for Security Monitoring and Testing
- ❖ Good for Flexible testing and debugging
 - Building distributed environments on a single machine
 - Kernel development and debugging
 - VM-based recording and replaying



Benefits of Virtualization

► Co-Existence of Different OSes on a single device

- ❖ Co-Existence of collaborative Apps requiring difference OS
- ❖ Good for Embedded Systems requiring different requirements of SWs

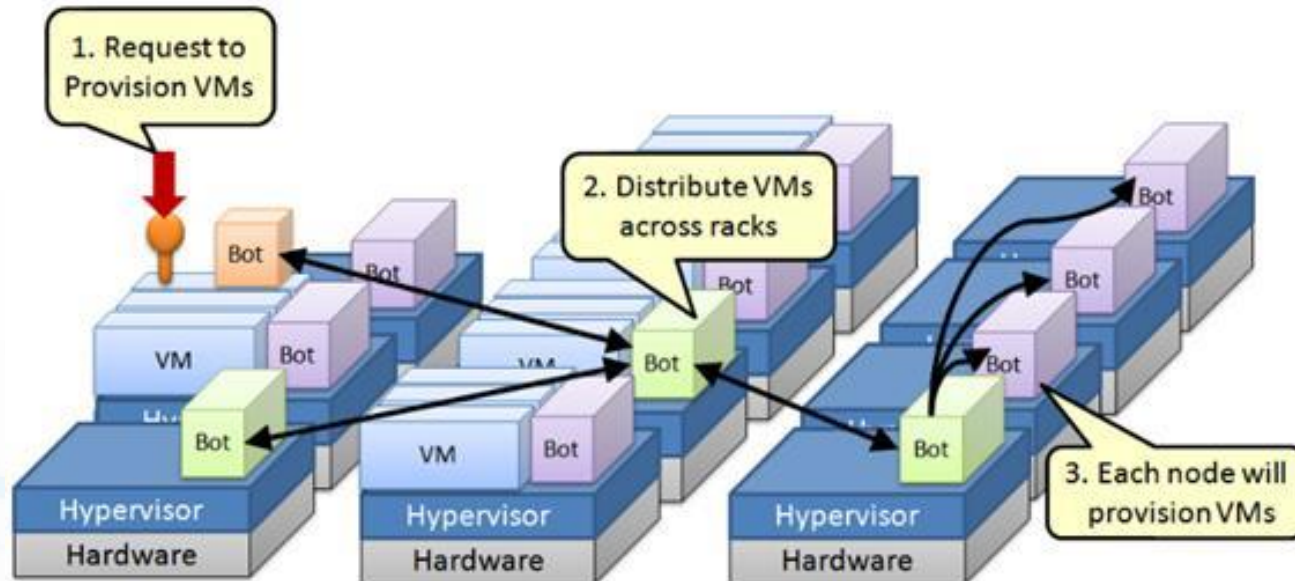


Virtualizing Real-Time Operating Systems with Windows

Benefits of Virtualization

► Fast and Easy Provisioning for Elastic Business

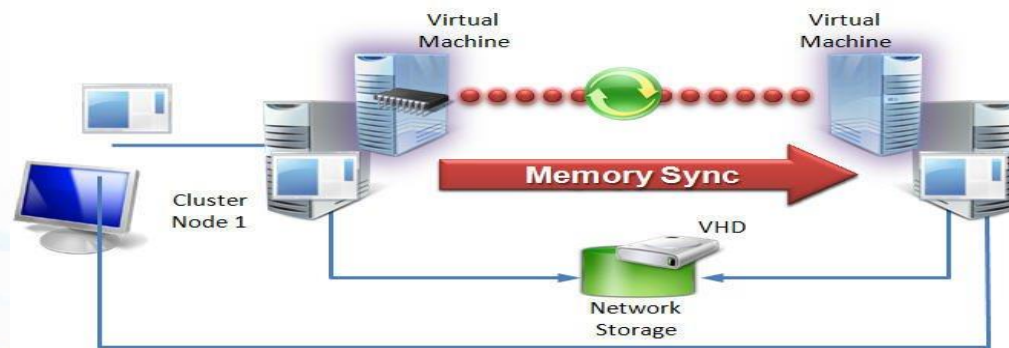
- ❖ quickly clone a gold image, master template, or existing virtual machine to get a server up and running within minutes.
- ❖ Automated Provisioning Computing Environment



Benefits of Virtualization

► Live VM migration

- ❖ Live migration: moving a running virtual machine or application between different physical machines without disconnecting the client or application
- ❖ Memory, storage, and network connectivity of the virtual machine are transferred from the original guest machine to the destination
- ❖ Good for **Load Balancing, High Availability**
- ❖ Good for **Uptime Optimization, disaster recovery**



Benefits of Virtualization

▶ VM is a SW

- ❖ can snapshot, freeze, **suspend, move, resume, clone**

▶ Templating

- ❖ create an OS + application VM, provide it to customers, use it to **create multiple instances** of that combination

▶ Encapsulation

- ❖ **All information** on virtual servers including boot disks is **saved in file format**

▶ Hardware-Independence

- ❖ Virtual servers run as is after **migration to different HW platform**
- ❖ Reducing hardware vendor lock-in

▶ Extend the life of older applications

Usages of Virtualization

Cloud computing
Virtual desktop infrastructure
Mobile virtualization

Virtualization & Cloud Computing

► Virtualization is the **core technology of cloud computing**

❖ Virtualization technology supports flexible VM based resource management which makes building Virtualized Data Center becomes possible

➤ Many IaaS solutions are built based on Commercial and Open-Source VMMs

■ E.g. Amazon Web Service, MS Azure, OpenStack, etc.

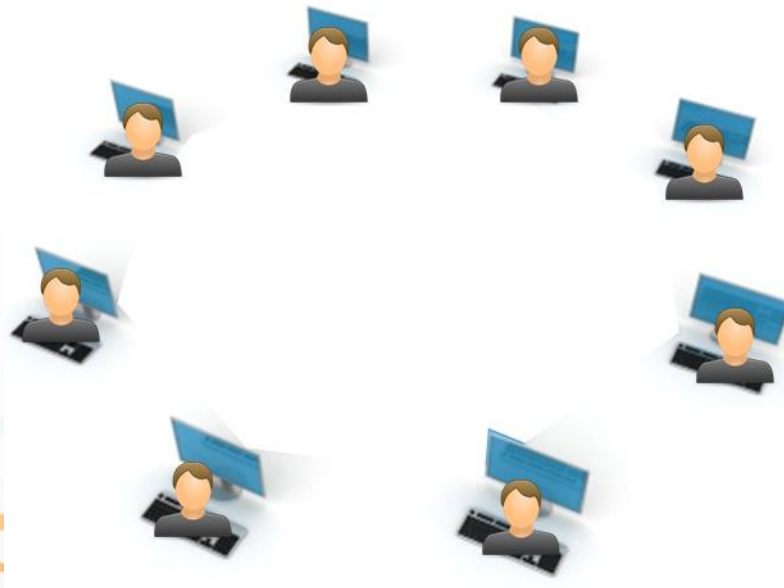
		Geographic Presence	Access Approach	VMM	Guest OS
Commercial IaaS Solution	Amazon EC2	Global	SSH, RDP	Xen	Linux, Windows
	MS Azure	Global	SSH, RDP	Hyper-V	Linux, Windows
	Flexiscale	UK	SSH	Xen	Linux, Windows
	GoGrid		SSH	Xen	Linux, Windows
	Joyent Cloud	US	SSH, VirtualMin	OS Level (Solaris Containers)	Open Solaris
	Rackspace Cloud Servers	US	SSH	Xen	Linux
Open Source IaaS Solution	OpenStack	-	SSH, RDP	Xen, Kvm	Linux, Windows
	CloudStack	-	SSH, RDP	Xen, Kvm	Linux, Windows
	OpenNebula	-	SSH, RDP	Xen, Kvm	Linux, Windows

Virtual Desktop Infrastructure (VDI)

▶ Traditional Desktop Provisioning Approach

❖ All desktops and desktop supporting environment, such as monitors, keyboards, mice, CPU, storage, memory, etc., are dedicated

- Energy Waste
- Management Cost Too Expensive
- Not Safety

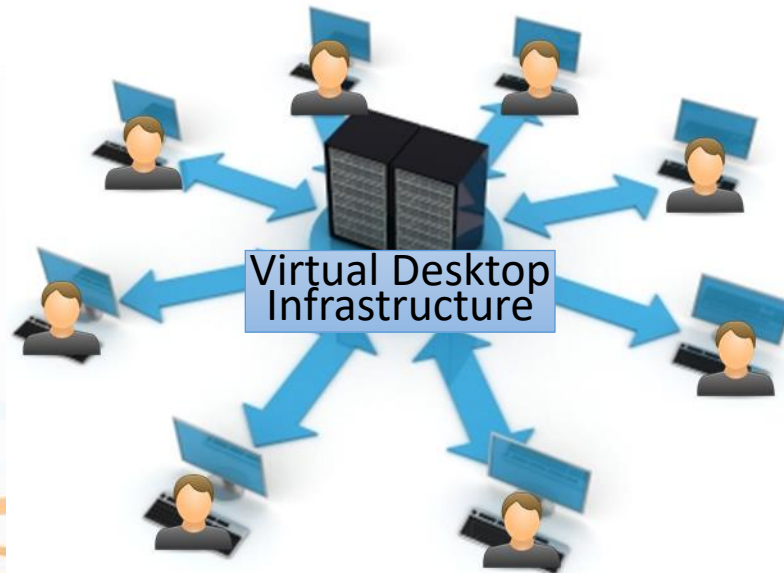


Virtual Desktop Infrastructure (VDI)

▶ VDI based Desktop Provisioning Approach

❖ Only necessary devices are dedicated and most of the hardware are shared by users

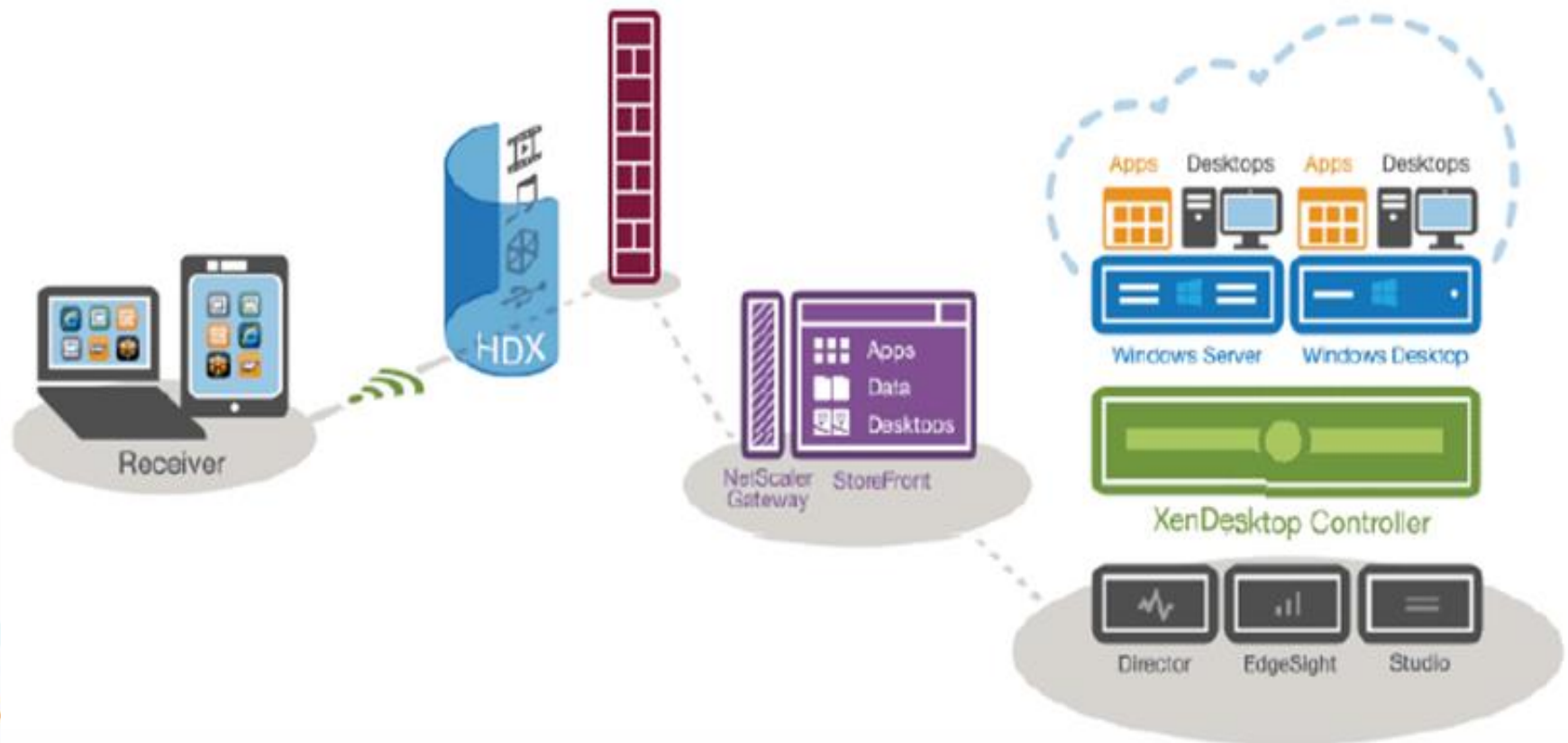
- Sufficient Energy Utilization
- Maintenance Cost Saving
- Centralized Resource Management



Virtual Desktop Infrastructure (VDI)

► VDI Example

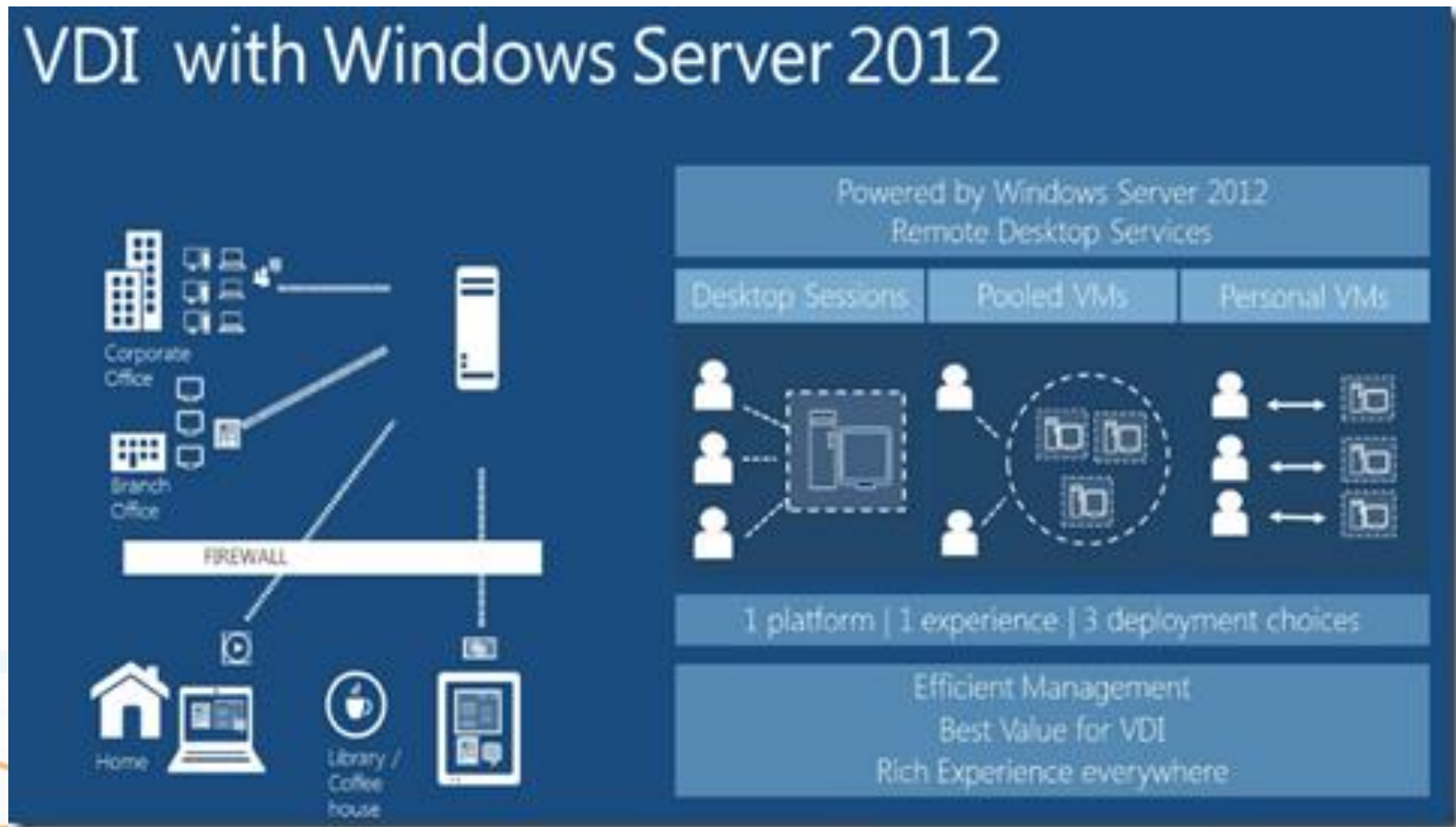
❖ Citrix XenDesktop



Virtual Desktop Infrastructure (VDI)

► VDI Example

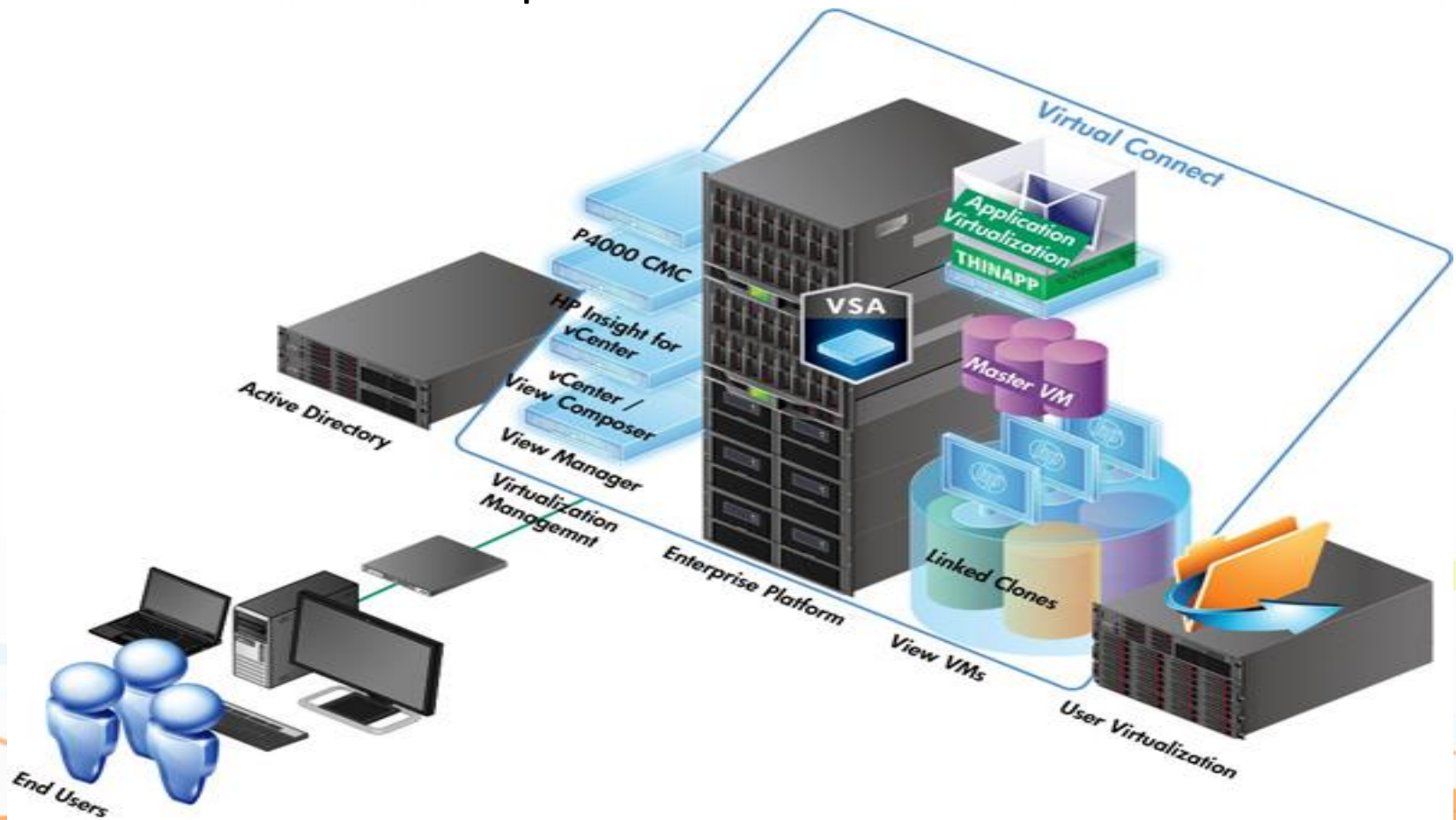
❖ Microsoft VDI



Virtual Desktop Infrastructure (VDI)

► VDI Example

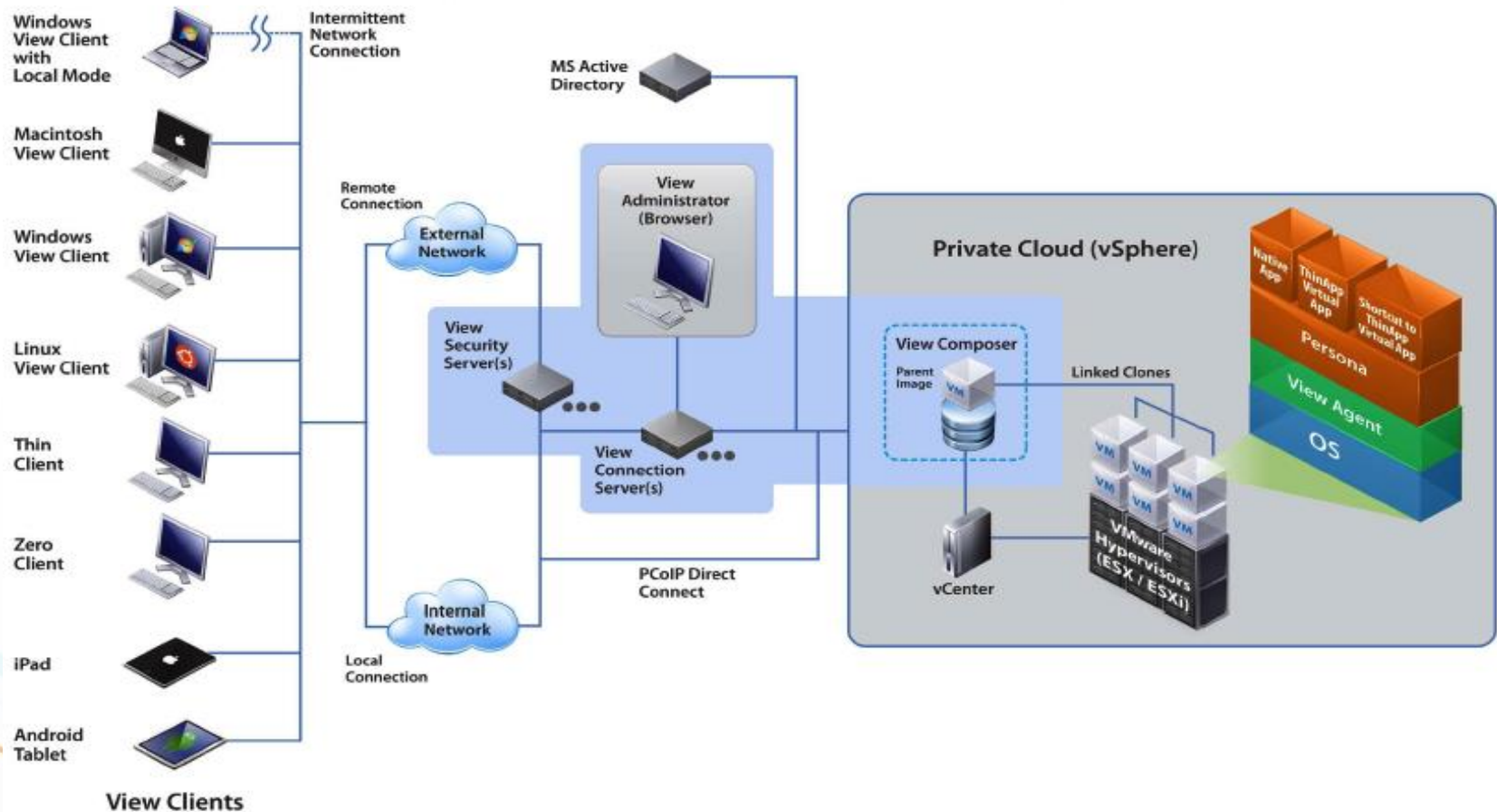
❖ HP Virtual Desktop



Virtual Desktop Infrastructure (VDI)

► VDI Example

❖ VMware View 5



Mobile Virtualization

► Trends of consumer electronics

❖ Digital convergence

➤ What do consumer electronics(CE) devices want to achieve?



Reliability

Trustworthiness for primary functions



Extensibility

Flexibility for adding features as needed

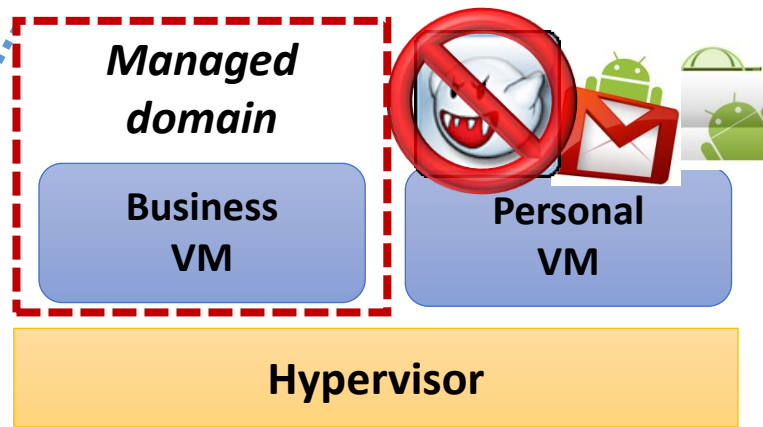
Business Phone



Personal Phone

Mobile Virtualization

- **BYOD: Bring Your Own Device**
 - ❖ Consolidating business and personal computing environments on a single device



Mobile Virtualization

▶ Mobile Virtualization Solutions

❖ Device Management Policy based Mobile Virtualization

- BlackBerry Balance from RIM
- MobileIron Virtual Smartphone Platform(VSP) from AT&T

❖ Hypervisor based Mobile Virtualization

- KVM on ARM
- XEN on ARM
- OKL4 Microvisor from Open Kernel Labs
- Motorola Evoke AQ4
- VMware Mobile Virtualization Platform (MVP)
- Red Bend: vLogix Mobile
- Cells
- Cellrox

모바일 보안, 대세는 클라우드 모바일 가상화

<http://platum.kr/archives/20204>

Mobile Virtualization

▶ Mobile Virtualization Solutions

❖ Container based Mobile Virtualization

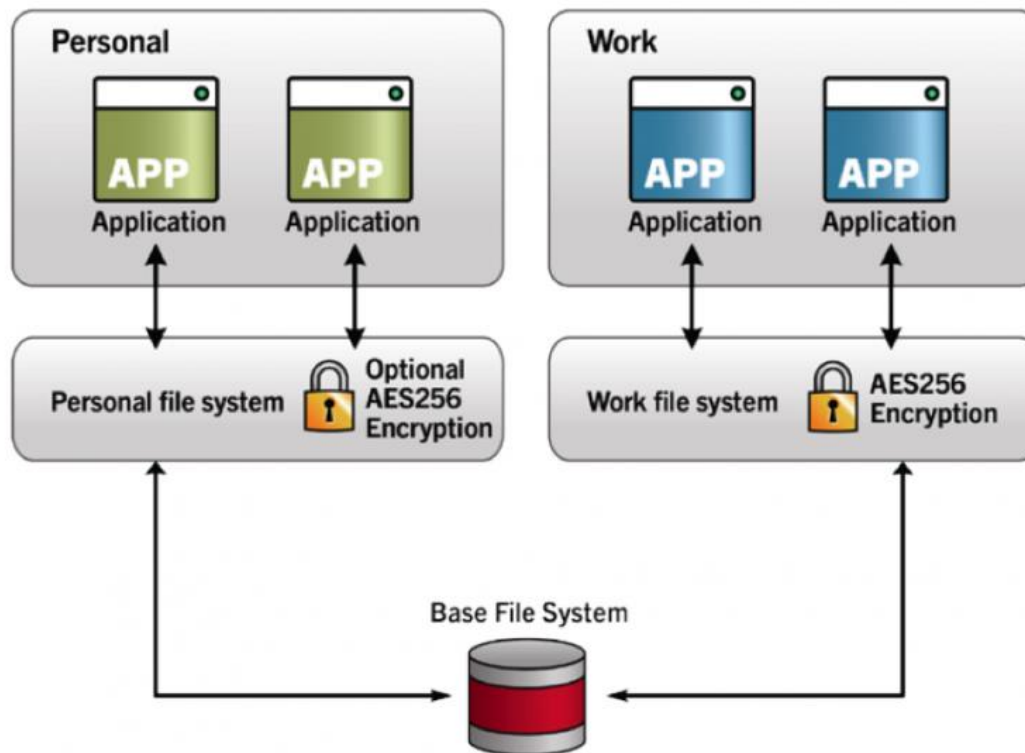
- Good Dynamics Technology
- Divide by Enterpoid
- TrustDroid
- Android 4.2: Multi-User

Mobile Virtualization Solution Example

▶ BlackBerry Balance from RIM

- ❖ Data Isolation
- ❖ Encryption

Cryptographically Partitioned File Systems



Mobile Virtualization Solution Example

► KVM on ARM

- ❖ CPU Virtualization
- ❖ Memory Virtualization
- ❖ I/O Virtualization
- ❖ Interrupt Virtualization
- ❖ Timer Virtualization

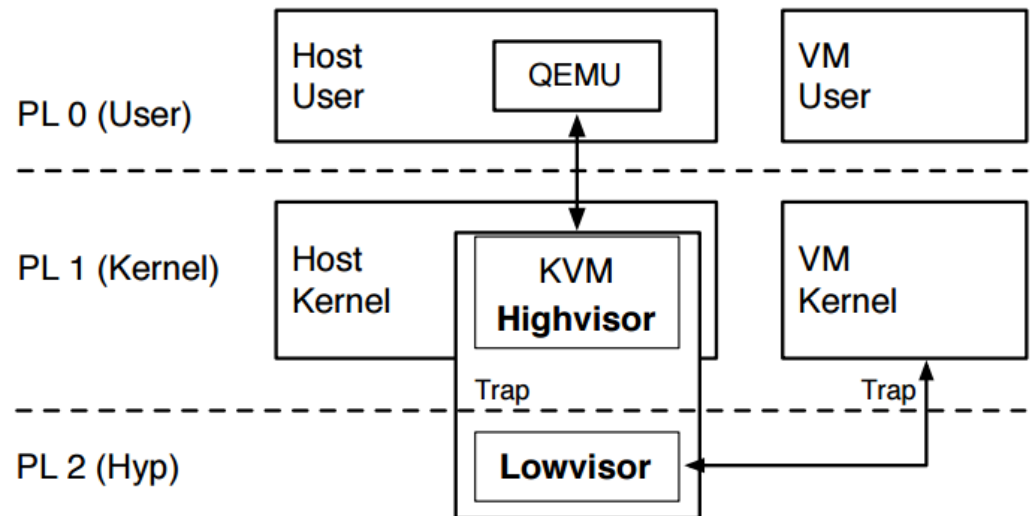


Figure 2: KVM/ARM System Architecture

Mobile Virtualization Solution Example

► VMware Mobile Virtualization Platform (MVP)



Fig. 2.5 MVP personal/enterprise screen shots [VMTN1]

Mobile Virtualization Solution Example

- ▶ Divide by Enterproid
 - ❖ Support for both Android and iOS
 - ❖ Dedicated password policy
 - ❖ Encryption
 - ❖ Data isolation
 - ❖ Clipboard restriction
 - ❖ Remote wipe



Fig. 2.7 Divide personal/enterprise screen shots

Summary

- ▶ What is Architecture Styles Related to Virtualization ?
 - ❖ Virtual Machine Architecture Style
- ▶ Implementation Levels of Virtualization
 - ❖ OS Level: Docker/LXC, HAL Level: Hypervisor
- ▶ What is Hardware Virtualization ?
 - ❖ CPU Virtualization, Memory Virtualization, I/O Virtualization
- ▶ VM and VMM(=Hypervisor)
 - ❖ Host Machine, Hypervisor(VMM), Guest Machine
- ▶ History of Virtualization
 - ❖ 1998 VMware Full Virtualization, 2003 XEN Para Virtualization, 2005 HW-assisted Virtualization
- ▶ Benefits of Virtualization
 - ❖ Optimize Utilization, Secured Co-Existence, Auto Provisioning, Live Migration
- ▶ Usages of Virtualization
 - ❖ Cloud Infra, VDI, Mobile Virtualization

Thank You!!



Any Questions ??